Making a Custom Composite Atlas with midbrain, cortex and subcortex

Me JUN 18

Hi Claude

Thanks for your quick edits on the VVIQ paper. it was really nice to get your expert eyes on it before submitting. Currently, it's submitted at Science Advances, but we will see where it ends up.

I have this idea for analyses that take more of a look at depression. I've been thinking about how to best model both structural and functional connectivity across cortical, subcortical, and midbrain regions, particularly as a way to explore depression-related pathways in the COST dataset. It's also something that could be valuable for exploring depression in the UK Biobank, which now has over 40,000 brain scans.

UK Biobank includes a number of parcellations and connectivity maps—for example, in this recent paper:
https://www.sciencedirect.com/science/article/pii/S105381192300558X?via%3Dihub

However, I've noticed that none of the included atlases appear to cover the midbrain, particularly key regions like the ventral tegmental area (VTA). I came across a recent preprint that investigates VTA in depression, and they make use of a "Levinson atlas" containing midbrain nuclei such as the VTA, DRN, and LC, but their atlas does not include cortical areas.

I haven't found an existing atlas that combines cortical, subcortical, and midbrain structures into a single framework that could support whole-brain connectomic analyses, both structurally and functionally. I may be fumbling around a bit due to inexperience in this area, but I see this as a potentially important step for modeling connectivity and predicting behavior or symptoms via links between midbrain, cortex, and subcortical circuits.
Is this something you think would be relatively straightforward, or would it be technically challenging or inadvisable to pursue? I'd really appreciate your perspective before going too far down the rabbit hole.

Best Regards
Timo Kvamme

Claude JUN 18

Hi Timo,

It _shouldn't_ be that difficult to add areas if the areas are distinct eg midbrain structures. I do not have a huge amount of experience doing it but assuming everything is in the same space and that the areas are not adjacent to one another then I do not imaging that it should be a major problem

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Me JUN 18

Thanks Claude.

I tried giving the question to chatgpt, and it said this:

✅ Step-by-step plan:

1. Ensure all atlases are in the same space and resolution

- Use **MNI152 2mm or 1mm space**—Levinson's atlas is in MNI152 1mm.
- Resample other atlases (e.g., Tian, Schaefer, Glasser) to **match the resolution and dimensions** of the Levinson atlas using `ANTs`, `FSL`, or `nilearn`.

```
# Example: Resample Tian to Levinson space using FSL
flirt -in tian.nii.gz -ref levinson_vta.nii.gz -out tian_resampled.nii.gz -applyxfm -usesqform
```

2. Relabel and separate parcels to avoid overlaps

- Make sure **each atlas has non-overlapping integer labels**.
- Assign new labels for Levinson ROIs (e.g. VTA = 701, DRN = 702, LC = 703, etc.).
- Keep Tian or cortical ROIs as-is, or relabel to a higher range (e.g., 1000+ for cortical).

Use `fslmaths`, `nilearn.image.math_img`, or `nibabel` in Python to modify labels.

3. Merge all atlases into one combined parcellation

You can use image math to sum non-overlapping labeled ROIs into one file.
Example in Python using `nibabel`:

```
in python:

import nibabel as nib
import numpy as np

lev = nib.load("levinson_vta.nii.gz")
tian = nib.load("tian_resampled.nii.gz")
schaefer = nib.load("schaefer_resampled.nii.gz")

combined = lev.get_fdata()
tian_data = tian.get_fdata()
schaefer_data = schaefer.get_fdata()

# Avoid label collisions
tian_data[tian_data > 0] += 100  # Offset to 100+
schaefer_data[schaefer_data > 0] += 1000  # Offset to 1000+

combined[tian_data > 0] = tian_data[tian_data > 0]
combined[schaefer_data > 0] = schaefer_data[schaefer_data > 0]

new_img = nib.Nifti1Image(combined, lev.affine)
nib.save(new_img, "combined_parcellation.nii.gz")
```

I'll try something like this, maybe it's not that difficult.

Best Regards
Timo Kvamme

Claude JUN 19

looks good but you may want to use a non-linear transform to ensure that they are all in the same space

Reminder returned JUL 17